

Francisco Gómez Serrano

PhD Candidate in Physics · Universidad de Chile

francisco.gomez.serrano96@gmail.com · [fran-gs-96.github.io](https://github.com/fran-gs-96) · [linkedin.com/in/francisco-gomez-serrano](https://www.linkedin.com/in/francisco-gomez-serrano)

RESEARCH INTERESTS

(Non-equilibrium) Statistical Physics, Theoretical Physics, Meteorology and Climatology.

EDUCATION

In progress	Ph.D. in Physics · <i>Universidad de Chile</i> <i>Thesis: active matter systems — fluidization and collective behavior in noisy, dense active systems.</i>
2023	M.Sc. in Meteorology and Climatology · <i>Universidad de Chile</i> <i>Summa Cum Laude</i>
2021	M.Sc. in Theoretical Physics · <i>Leiden University, The Netherlands</i> <i>Cum Laude</i>
2019	Professional Degree (Licenciatura) in Physics · <i>Pontificia Universidad Católica de Valparaíso</i> <i>Summa Cum Laude</i>

PUBLICATIONS

In prep.	“Characterization of synoptic-scale atmospheric patterns related to electric activity” (from M.Sc. thesis).
2025	Gómez, F., Pérez, P., Menares, C., & Flemming, Z. “Sulfur dioxide concentration forecasting using a deep learning model in Quintero, Chile.” <i>Atmospheric Pollution Research</i> . https://doi.org/10.1016/j.apr.2025.102534
2021	Astefanesei, D., Blázquez-Salcedo, J.L., Gómez, F., & Rojas, R. “Thermodynamically stable asymptotically flat hairy black holes with a dilaton potential: the general case.” <i>Journal of High Energy Physics</i> . https://doi.org/10.1007/JHEP02(2021)233
2019	Astefanesei, D., Choque, D., Gómez, F., & Rojas, R. “Thermodynamically stable asymptotically flat hairy black holes with a dilaton potential.” <i>Journal of High Energy Physics</i> . https://doi.org/10.1007/JHEP03(2019)205

TALKS & PRESENTATIONS

2026	“Explaining increasing fluidization in noisy-dense-active systems using renormalization group.” Poster presentation — TREFEMAC 2026, Bariloche.
2024	“Characterization of synoptic-scale patterns related to electric activity.” Oral presentation — 7th Congress of Physical Oceanography, Meteorology, and Climate of Southern East Pacific.
2023	“Characterization of synoptic-scale patterns related to electric activity.” Poster presentation — 1st Congress for Postgraduate Students, Universidad de Chile.
2023	“Non-linear time series forecasting using Deep Learning V.2.” Invited talk for users of South American High-Performance Computers.
2022	“Non-linear time series forecasting using Deep Learning.” Invited talk for users of the National Laboratory of High-Performance Computing.
2021	“Science beyond the lab.” Outreach talk, Liceo Juana Ross de Edwards.

TEACHING EXPERIENCE

2026	Lecturer, Physics of the Atmosphere · <i>Escuela Técnica Aeronáutica</i>
2024	University Lecturer, Physics · <i>Pontificia Universidad Católica de Valparaíso</i>
2015 – 2017	Teaching Assistant · <i>Pontificia Universidad Católica de Valparaíso</i> Assisted in several undergraduate courses in physics and mathematics.

GRANTS, SCHOLARSHIPS & AWARDS

2025	ANID Doctorate Scholarship — merit-based scholarship supporting doctoral studies at national universities.
2019	Becas Chile (ANID) — merit-based scholarship supporting studies at top international universities.
2018	Alumni Award, PUCV — recognition for outstanding commitment to studies and university life.
2013	Best Mathematician, High School — recognition for outstanding aptitude in mathematics.

OTHER PROFESSIONAL EXPERIENCE

2024 – 2025	Founding Partner · <i>AiQlim Intelligence SpA</i>
2023 – 2024	Data Scientist & Project Engineer · <i>DUAM SA</i> Adapted NLP models to assess web-search strategies and automate classification tasks; contributed to strategy/innovation roadmaps.
2023	Researcher · <i>CR2 – Center for Climate and Resilience Research</i> Characterized synoptic-scale atmospheric patterns related to electric activity.
2023	Freelance Research Consultant · <i>International Maize and Wheat Improvement Center (CIMMYT)</i> Built an ETL pipeline for agrometeorological data.
2023	Freelance Consultant · <i>DUAM SA</i> Automated data collection on topics of interest from multiple webpages.

TECHNICAL SKILLS & LANGUAGES

Languages: Spanish (native), English (professional working proficiency).
Programming: Python, MATLAB.
Generative AI: proficient use of AI-assisted tools (e.g., Claude, Gemini, etc.) as a research and coding assistant.

OTHER

Science outreach volunteer with Explora (ANID) and Fundación Conectado Aprendo (2022).
Marathon runner and Taekwondo black belt (2nd Dan).